

Calibration-Aware Recurrent Neural Networks for Multivariate Probabilistic Forecasting under Distribution Shift

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Abstract

Probabilistic forecasts are useful only when their stated uncertainty is credible. Likelihood-trained recurrent neural networks may be miscalibrated, while Bayesian model averaging is not guaranteed to remain calibrated when the likelihood, prior, or conditional dynamics are misspecified. This paper develops the Calibration-Aware Recurrent Neural Network (CA-RNN), adapting the calibration-aware frequentist and Bayesian learning framework of Huang, Park, and Simeone to continuous multivariate forecasting. The proposed training criterion augments joint negative log likelihood with a differentiable discrepancy between projected probability integral transforms (PITs) and the uniform distribution. A sequential extension additionally penalizes lagged dependence in the projected PIT sequence. The empirical design compares RNN, Bayesian RNN, CA-RNN, and calibration-aware Bayesian RNN, and then extends the comparison to model misspecification and distribution shift. In the four-asset application, sequential CA-RNN reduces projected-PIT calibration error consistently across optimization seeds and chronological market regimes, but does not uniformly improve point or proper-score accuracy. The results identify a regime-dependent calibration–accuracy tradeoff rather than universal dominance.

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1 Introduction

1.1 Motivation

A point forecast reports one possible future; a probabilistic forecast reports a distribution of possible futures. The latter is operationally meaningful only if events assigned probability α occur at approximately rate α , while the distribution remains sufficiently sharp to discriminate among outcomes. Modern neural forecasters can optimize likelihood successfully without achieving reliable finite-sample calibration. Bayesian learning represents parameter uncertainty, but its calibration guarantees depend on idealized assumptions about the likelihood, prior, posterior computation, and data-generating mechanism.

Huang, Park, and Simeone propose calibration-aware Bayesian neural networks (CA-BNNs), combining a data-dependent calibration regularizer with the data-independent KL regularizer of variational Bayesian learning [1]. Their experiments compare frequentist, Bayesian, calibration-aware frequentist, and calibration-aware Bayesian classifiers; vary the calibration weight; and report expected calibration error and reliability diagrams. Their conclusion identifies evaluation under distribution shift as a direction for future work. This paper transfers that methodological template to continuous multivariate sequential prediction and makes misspecification and distribution shift a central part of the evaluation.

1.2 Research questions

The primary research question is:

Can calibration-aware training improve multivariate probabilistic RNN forecasts, and does that improvement persist under model misspecification and distribution shift without an unacceptable loss of sharpness or predictive accuracy?

The supporting questions are:

- RQ1. How does the calibration weight affect projected calibration, proper predictive scores, sharpness, and point accuracy?
- RQ2. Does calibration-aware Bayesian learning behave differently from its frequentist counterpart for dependent sequential data?
- RQ3. Does penalizing serial dependence in projected PIT values improve sequential forecast adequacy beyond marginal PIT uniformity?
- RQ4. How do the four model classes deteriorate under changes in volatility, dependence, tail behavior, conditional dynamics, and structural regime?

1.3 Contributions

The contributions are methodological and empirical:

1. a continuous-outcome analogue of calibration-aware neural training based on differentiable projected PIT discrepancies;
2. an extension targeting temporal dependence in forecast PIT sequences;
3. a variational Bayesian recurrent formulation combining calibration and KL regularization;

4. a matched four-model experiment that parallels [1]; and
5. a controlled study of calibration degradation under misspecification and distribution shift, including a multivariate financial application.

1.4 Scope and non-claims

Finite projected calibration is weaker than complete multivariate distributional calibration. Calibration of a finite test set is also not a guarantee of conditional calibration at every history. We therefore avoid claims that a small collection of projections proves joint correctness. The proposed regularizer is an explicit, diagnostically aligned training objective whose value must be established empirically with proper scores and out-of-sample tests.

2 Related Work

2.1 Calibration-aware Bayesian learning

The organizing reference is [1]. It distinguishes data-dependent calibration regularization from the data-independent prior regularization used by variational Bayesian inference. Its CA-FNN objective adds a differentiable approximation of expected calibration error to cross-entropy, and its CA-BNN objective places the same calibration penalty inside the variational expectation while retaining a KL penalty. The paper uses Gaussian variational parameters and the reparameterization trick, and its main comparison comprises FNN, BNN, CA-FNN, and CA-BNN. The reported experiments emphasize calibration-weight sweeps and reliability diagrams.

2.2 Broader literature review to be completed

This section is intentionally a structured stub. It will cover:

1. probabilistic calibration, auto-calibration, and PIT diagnostics;
2. proper scoring rules and calibration-sharpness decompositions;
3. multivariate calibration, copula diagnostics, and projection methods;
4. recurrent probabilistic forecasting and Bayesian recurrent networks;
5. variational inference under misspecification;
6. calibration under temporal dependence and distribution shift; and
7. probabilistic forecasting and risk evaluation in financial time series.

Only sources actually reviewed will be added to the bibliography. The local literature archive is maintained in `docs/refs/`.

3 Forecasting Setup and Calibration

3.1 Sequential prediction problem

Let $(Y_t)_{t=1}^T$ be an adapted stochastic process with $Y_t \in \mathbb{R}^d$ and filtration $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$. Given a history length H , define $X_t = (Y_{t-H+1}, \dots, Y_t)$. The task is to estimate a conditional predictive

distribution

$$P_t(\cdot) = \mathbb{P}(Y_{t+1} \in \cdot \mid \mathcal{F}_t)$$

with a model $P_{\theta,t}$. Training, validation, and test blocks are ordered chronologically. Standardization parameters are estimated on the training block only.

3.2 Joint Gaussian predictive head

For the initial specification, the predictive law is

$$Y_{t+1} \mid \mathcal{F}_t, \theta \sim \mathcal{N}_d(\mu_{\theta,t}, \Sigma_{\theta,t}), \quad \Sigma_{\theta,t} = L_{\theta,t} L_{\theta,t}^\top. \quad (1)$$

The network outputs the d components of $\mu_{\theta,t}$ and the $d(d+1)/2$ lower-triangular entries of $L_{\theta,t}$. An unconstrained diagonal output $s_{t,j}$ is transformed as

$$L_{t,jj} = \text{softplus}(s_{t,j}) + \varepsilon, \quad \varepsilon > 0.$$

Proposition 3.1 (Validity of the predictive covariance). *For every network output, $\Sigma_{\theta,t} = L_{\theta,t} L_{\theta,t}^\top$ is symmetric positive definite.*

Proof. Symmetry follows from transposition. Since every diagonal entry of the triangular matrix $L_{\theta,t}$ is strictly positive, $L_{\theta,t}$ is nonsingular. Hence, for any nonzero $x \in \mathbb{R}^d$,

$$x^\top \Sigma_{\theta,t} x = x^\top L_{\theta,t} L_{\theta,t}^\top x = \|L_{\theta,t}^\top x\|_2^2 > 0.$$

Thus $\Sigma_{\theta,t}$ is positive definite. □

3.3 Negative log likelihood

For an index set I , the mean joint Gaussian negative log likelihood is

$$\mathcal{L}_{\text{NLL}}(\theta; I) = -\frac{1}{|I|} \sum_{t \in I} \log p_\theta(Y_{t+1} \mid X_t) \quad (2)$$

$$= \frac{1}{2|I|} \sum_{t \in I} \left[d \log(2\pi) + 2 \sum_{j=1}^d \log L_{t,jj} + e_t^\top \Sigma_{\theta,t}^{-1} e_t \right], \quad (3)$$

where $e_t = Y_{t+1} - \mu_{\theta,t}$. Computation uses triangular solves rather than an explicit matrix inverse.

Proposition 3.2 (Strict propriety of the population log score). *Let $P_0(\cdot \mid \mathcal{F}_t)$ have density p_0 and let p be any candidate conditional density with common support. Then*

$$\mathbb{E}_{P_0}[-\log p(Y_{t+1} \mid \mathcal{F}_t)] - \mathbb{E}_{P_0}[-\log p_0(Y_{t+1} \mid \mathcal{F}_t)] = \mathbb{E}[\text{KL}(P_0(\cdot \mid \mathcal{F}_t) \| P(\cdot \mid \mathcal{F}_t))] \geq 0.$$

Equality holds only when the conditional densities agree almost surely.

Proof. Condition on $\mathcal{F}_t$, subtract the two expected log scores, and recognize the conditional KL divergence. Nonnegativity follows from Gibbs' inequality. Taking expectations over $\mathcal{F}_t$ proves the result. □

This proposition explains why the calibration penalty must supplement rather than replace a proper score: calibration alone can admit uninformative forecasts.

3.4 Projected probability integral transforms

Fix unit vectors $a_1, \dots, a_R \in \mathbb{R}^d$. In the experiments they comprise the d coordinate directions, one equal-weight direction, and four random unit directions fixed by a documented seed. Define

$$U_{t,r} = F_{\theta,t,r}(a_r^\top Y_{t+1}), \quad (4)$$

where $F_{\theta,t,r}$ is the predictive CDF of $a_r^\top Y_{t+1}$.

Proposition 3.3 (Projected Gaussian law). *Under (1),*

$$a_r^\top Y_{t+1} \mid \mathcal{F}_t, \theta \sim \mathcal{N}\left(a_r^\top \mu_{\theta,t}, a_r^\top \Sigma_{\theta,t} a_r\right),$$

and therefore

$$U_{t,r} = \Phi\left(\frac{a_r^\top (Y_{t+1} - \mu_{\theta,t})}{\sqrt{a_r^\top \Sigma_{\theta,t} a_r}}\right).$$

Proof. A linear transformation of a multivariate Gaussian is Gaussian. Its mean and variance follow from linearity of expectation and the covariance transformation rule. Applying its CDF gives the displayed PIT. $\square$

Theorem 3.4 (Sequential PIT property). *Suppose $F_{0,t,r}$ is the true, continuous conditional CDF of $a_r^\top Y_{t+1}$ given $\mathcal{F}_t$. Then $U_{t,r} = F_{0,t,r}(a_r^\top Y_{t+1})$ is uniformly distributed conditional on $\mathcal{F}_t$. For a fixed r , the sequence $(U_{t,r})_t$ is independent and $\text{Uniform}(0, 1)$.*

Proof. For $u \in [0, 1]$, continuity and the conditional probability integral transform give

$$\mathbb{P}(U_{t,r} \leq u \mid \mathcal{F}_t) = u.$$

Thus $U_{t,r}$ is conditionally uniform and its conditional law does not depend on $\mathcal{F}_t$. For times $t_1 < \dots < t_m$, iterated conditioning yields

$$\begin{aligned} \mathbb{P}(U_{t_1,r} \leq u_1, \dots, U_{t_m,r} \leq u_m) &= \mathbb{E}\left[\mathbf{1}_{\{U_{t_1,r} \leq u_1, \dots, U_{t_{m-1},r} \leq u_{m-1}\}} \mathbb{P}(U_{t_m,r} \leq u_m \mid \mathcal{F}_{t_m})\right] \\ &= u_m \mathbb{P}(U_{t_1,r} \leq u_1, \dots, U_{t_{m-1},r} \leq u_{m-1}). \end{aligned}$$

Induction gives $\prod_{j=1}^m u_j$, proving independence and uniformity. $\square$

Remark 3.5. *Uniformity for finitely many a_r is necessary but not sufficient for equality of multivariate distributions. If all one-dimensional projections agree, the Cramér–Wold device identifies the joint distribution; a finite projection set only provides a tractable approximation and diagnostic.*

Training uses the analytic Gaussian PIT above. Out-of-sample evaluation instead uses the common sample-rank approximation

$$\hat{U}_{t,r} = \frac{\sum_{s=1}^S \mathbf{1}\{a_r^\top Y_{t+1}^{(s)} < a_r^\top Y_{t+1}\} + 1/2}{S + 1},$$

with $S = 500$ predictive draws. This supports the same diagnostic for deterministic Gaussian forecasts and Bayesian posterior-predictive mixtures, at the cost of finite-simulation error.

4 Calibration-Aware RNN Methodology

4.1 Recurrent architecture

For a single-layer GRU, with elementwise product $\odot$,

$$\begin{aligned} r_t &= \sigma(W_{ir}Y_t + b_{ir} + W_{hr}h_{t-1} + b_{hr}), \\ z_t &= \sigma(W_{iz}Y_t + b_{iz} + W_{hz}h_{t-1} + b_{hz}), \\ n_t &= \tanh(W_{in}Y_t + b_{in} + r_t \odot (W_{hn}h_{t-1} + b_{hn})), \\ h_t &= n_t + z_t \odot (h_{t-1} - n_t). \end{aligned}$$

Linear heads map the final hidden state to $\mu_{\theta,t}$ and the unconstrained entries of $L_{\theta,t}$.

4.2 Differentiable projected calibration loss

Let $q_g = g/(G+1)$ for $g = 1, \dots, G$. The empirical CDF of PIT values is nondifferentiable in model parameters because it uses indicators. Replace the indicator $\mathbf{1}\{U_{t,r} \leq q_g\}$ by

$$s_\tau(q_g - U_{t,r}) = \sigma\left(\frac{q_g - U_{t,r}}{\tau}\right), \quad \tau > 0.$$

For a consecutive training batch B , define

$$\mathcal{L}_{\text{cal}}(\theta; B) = \frac{1}{GR} \sum_{g=1}^G \sum_{r=1}^R \left[\frac{1}{|B|} \sum_{t \in B} s_\tau(q_g - U_{t,r}) - q_g \right]^2. \quad (5)$$

This is a grid approximation to an integrated squared CDF discrepancy, analogous in purpose—but not in definition—to the differentiable AECE used for classification in [1].

Proposition 4.1 (Differentiability). *If the network outputs are differentiable in θ and $a_r^\top \Sigma_{\theta,t} a_r > 0$, then $\mathcal{L}_{\text{cal}}(\theta; B)$ is differentiable in θ .*

Proof. The Cholesky parameterization, affine projection, positive square root, Gaussian CDF, logistic smoother, finite sums, and squaring operations are differentiable on the stated domain. Their composition is therefore differentiable. $\square$

Proposition 4.2 (Zero-temperature limit). *Fix PIT values such that $U_{t,r} \neq q_g$ for every t, r, g . As $\tau \downarrow 0$, (5) converges to the squared grid discrepancy formed with the empirical PIT CDF.*

Proof. For nonzero x , $\sigma(x/\tau) \rightarrow \mathbf{1}\{x > 0\}$ as $\tau \downarrow 0$. Every sum is finite, so the limit passes through the averages and squares. The exclusion of ties removes the ambiguous boundary value. $\square$

4.3 Sequential calibration loss

PIT uniformity does not rule out forecastable temporal structure. With $\tilde{U}_{t,r} = U_{t,r} - 1/2$ and maximum lag K , define

$$\mathcal{L}_{\text{seq}}(\theta; B) = \frac{1}{RK} \sum_{r=1}^R \sum_{k=1}^K \left[\frac{1}{|B| - k} \sum_{t \in B: k \text{ available}} \tilde{U}_{t,r} \tilde{U}_{t-k,r} \right]^2. \quad (6)$$

Under the ideal conditional forecast, the population quantities targeted by (6) are zero by the sequential PIT theorem. The converse is not true: finitely many zero autocovariances do not establish independence.

4.4 Frequentist CA-RNN objective

The proposed frequentist objective is

$$\hat{\theta}_{\text{CA}} = \arg \min_{\theta} \{ \mathcal{L}_{\text{NLL}}(\theta; \mathcal{D}) + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}(\theta; \mathcal{D}) + \lambda_{\text{seq}} \mathcal{L}_{\text{seq}}(\theta; \mathcal{D}) \}. \quad (7)$$

The replication experiment initially fixes $\lambda_{\text{seq}} = 0$, so that (7) follows the source paper’s task-loss-plus-calibration structure. The sequential term is evaluated separately as a proposed extension.

4.5 Bayesian CA-RNN objective

Let every scalar neural parameter have variational posterior and prior

$$q_{\phi}(\theta_j) = \mathcal{N}(\mu_j, \sigma_j^2), \quad p(\theta_j) = \mathcal{N}(0, \sigma_0^2), \quad \sigma_j = \text{softplus}(\rho_j) + \epsilon.$$

Sampling uses the reparameterization

$$\theta_j = \mu_j + \sigma_j \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1).$$

The independent-Gaussian KL term has closed form

$$\text{KL}(q_{\phi} \| p) = \sum_j \left[\log \frac{\sigma_0}{\sigma_j} + \frac{\sigma_j^2 + \mu_j^2}{2\sigma_0^2} - \frac{1}{2} \right]. \quad (8)$$

The calibration-aware variational objective is

$$\mathcal{F}^{\text{CA}}(\phi) = \mathbb{E}_{\theta \sim q_{\phi}} [\mathcal{L}_{\text{NLL}}(\theta; \mathcal{D}) + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}(\theta; \mathcal{D}) + \lambda_{\text{seq}} \mathcal{L}_{\text{seq}}(\theta; \mathcal{D})] \quad (9)$$

$$+ \frac{\beta}{N} \text{KL}(q_{\phi}(\theta) \| p(\theta)). \quad (10)$$

This is the sequential continuous-outcome analogue of the CA-BNN free energy in [1]. The factor $1/N$ matches the mean per-observation likelihood convention. The weight β remains a validation-selected generalized-Bayes hyperparameter.

Proposition 4.3 (Unbiased reparameterized gradient for the expectation). *Assume differentiation and expectation may be interchanged. If $\theta^{(m)} = \mu + \sigma(\rho) \odot \varepsilon^{(m)}$ with independent $\varepsilon^{(m)} \sim \mathcal{N}(0, I)$, then the Monte Carlo gradient of the expected data-dependent part of (10) is unbiased.*

Proof. Write the expectation with respect to the parameter-free distribution of ε . Under the interchange assumption,

$$\nabla_{\phi} \mathbb{E}_{\varepsilon} [f(\mu + \sigma(\rho) \odot \varepsilon)] = \mathbb{E}_{\varepsilon} \nabla_{\phi} f(\mu + \sigma(\rho) \odot \varepsilon).$$

The sample average of independent terms has this expectation. The KL gradient is computed analytically from (8). $\square$

Remark 4.4 (Dependent mini-batches). *Unlike independent classification observations, overlapping time-series windows are dependent. Consecutive batches are required to evaluate $\mathcal{L}_{\text{seq}}$, but their nonlinear calibration loss is not asserted to be an unbiased estimate of the full-sample calibration loss. Batch length and history overlap are therefore methodological hyperparameters and sensitivity targets.*

In the implementation, both calibration penalties are batch-local: lag pairs crossing mini-batch boundaries are omitted. Epoch summaries average batch means, so the final shorter batch receives the same summary weight as a full batch. The displayed dataset-level objectives are the target criteria; stochastic optimization uses this documented batch-local surrogate.

4.6 Training algorithms

Algorithm 1 Frequentist CA-RNN training

Require: Ordered observations $\mathcal{D}$; history H ; projections A ; weights $\lambda_{\text{cal}}, \lambda_{\text{seq}}$; epoch limit E ; gradient threshold c

Ensure: Fitted parameters $\hat{\theta}$ selected without test information

```

1: Split  $\mathcal{D}$  chronologically into  $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{te}}$ 
2: Fit standardization on  $\mathcal{D}_{\text{tr}}$  and construct length- $H$  windows
3: Initialize  $\theta$  and fix  $A = (a_1, \dots, a_R)^\top$ 
4: for  $e = 1, \dots, E$  do
5:   for consecutive mini-batch  $B \subset \mathcal{D}_{\text{tr}}$  do
6:      $(\mu_{\theta,t}, L_{\theta,t})_{t \in B} \leftarrow \text{CARNNFORWARD}(B, \theta)$ 
7:     Compute  $\mathcal{L}_{\text{NLL}}$  by (3) and projected PITs by (4)
8:     Compute  $\mathcal{L}_{\text{cal}}$  by (5) and  $\mathcal{L}_{\text{seq}}$  by (6)
9:      $\mathcal{J} \leftarrow \mathcal{L}_{\text{NLL}} + \lambda_{\text{cal}}\mathcal{L}_{\text{cal}} + \lambda_{\text{seq}}\mathcal{L}_{\text{seq}}$ 
10:     $g \leftarrow \text{CLIPGRADNORM}(\nabla_{\theta}\mathcal{J}, c)$ 
11:    Update  $\theta$  with AdamW using  $g$ 
12:  end for
13:  Evaluate the prespecified selection score on  $\mathcal{D}_{\text{val}}$ 
14:  Save  $\theta$  if the validation score improves
15:  if the patience criterion is met then
16:    break
17:  end if
18: end for
19: Restore the selected checkpoint  $\hat{\theta}$ 
20: Evaluate  $\hat{\theta}$  once on  $\mathcal{D}_{\text{te}}$ 

```

For fixed-budget causal loss ablations, checkpoint selection and the patience test in Algorithm 1 are disabled, and all specifications are compared after the same number of epochs.

Algorithm 2 is the recurrent continuous-outcome adaptation of the CA-BNN procedure in [1].

5 Model Misspecification and Distribution Shift

5.1 Distinguishing misspecification from shift

Model misspecification occurs when the true conditional law $P_0(Y_{t+1} \mid \mathcal{F}_t)$ is outside the fitted family, even if the data law is stationary. Distribution shift occurs when the evaluation law differs from the development law. The two can coexist: a Gaussian predictive head is misspecified under heavy-tailed innovations, while a transition from Gaussian training innovations to heavy-tailed test innovations is also a distribution shift.

Let P_0 denote the reference law and let $P_{\delta,s}$ denote stress family s at severity $\delta \geq 0$, with $P_{0,s} = P_0$. For metric M , define degradation

$$\Delta_M^{(m,s)}(\delta) = M(P_{\delta,s}, \hat{P}_m) - M(P_{0,s}, \hat{P}_m), \quad (11)$$

where m indexes the forecasting method. For metrics in which smaller is better, positive Δ_M denotes deterioration. Coverage is reported as absolute error from the nominal level before differencing.

Algorithm 2 Variational calibration-aware Bayesian RNN training

Require: Ordered observations $\mathcal{D}$; projections A ; prior scale σ_0 ; Monte Carlo count M ; loss weights $\lambda_{\text{cal}}, \lambda_{\text{seq}}, \beta$; epoch limit E

Ensure: Variational parameters $\hat{\phi} = (\hat{\mu}, \hat{\rho})$

- 1: Construct chronological splits and training-only standardization as in Algorithm 1
 - 2: Initialize $\phi = (\mu, \rho)$ and set $\sigma(\rho) = \text{softplus}(\rho) + \epsilon$
 - 3: **for** $e = 1, \dots, E$ **do**
 - 4: **for** consecutive mini-batch $B \subset \mathcal{D}_{\text{tr}}$ **do**
 - 5: **for** $m = 1, \dots, M$ **do**
 - 6: Draw $\varepsilon^{(m)} \sim \mathcal{N}(0, I)$ and set $\theta^{(m)} \leftarrow \mu + \sigma(\rho) \odot \varepsilon^{(m)}$
 - 7: Use $\theta^{(m)}$ consistently at every recurrent time step
 - 8: Compute $\mathcal{J}_{\text{data}}^{(m)} \leftarrow \mathcal{L}_{\text{NLL}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}^{(m)} + \lambda_{\text{seq}} \mathcal{L}_{\text{seq}}^{(m)}$
 - 9: **end for**
 - 10: $\hat{\mathcal{F}}^{\text{CA}} \leftarrow M^{-1} \sum_{m=1}^M \mathcal{J}_{\text{data}}^{(m)} + (\beta/N) \text{KL}(q_{\phi} \| p)$
 - 11: Differentiate $\hat{\mathcal{F}}^{\text{CA}}$, clip its gradient, and update ϕ with AdamW
 - 12: **end for**
 - 13: Evaluate validation NLL at posterior-mean weights and update the selected checkpoint
 - 14: **if** the patience criterion is met **then**
 - 15: **break**
 - 16: **end if**
 - 17: **end for**
 - 18: Restore $\hat{\phi}$
 - 19: **for** each test origin t and predictive draw s **do**
 - 20: Draw $\theta^{(s)} \sim q_{\hat{\phi}}$ and then $Y_{t+1}^{(s)} \sim p_{\theta^{(s)}}(\cdot | \mathcal{F}_t)$
 - 21: **end for**
 - 22: Evaluate the posterior predictive mixture on $\mathcal{D}_{\text{te}}$
-

5.2 Stress families

The following are prospective isolated stress families for a confirmatory study; they were not all executed factorially in the present experiments:

1. **Scale shift:** innovation covariance is multiplied by a severity function while dependence is held fixed.
2. **Dependence shift:** correlations or a dynamic correlation state change while marginal variances are controlled.
3. **Tail shift:** Gaussian innovations transition to Student- t , mixtures, or asymmetric downside contamination.
4. **Volatility misspecification:** ARCH/GARCH or stochastic-volatility dynamics violate the Gaussian head's conditionally homoskedastic approximation.
5. **Mean-dynamics shift:** nonlinear interactions or an abrupt change in recurrent coefficients alter the conditional location.

The implemented developmental curve uses common random numbers and jointly changes scale and common correlation. Models are selected without examining its stressed test metrics and

are frozen after fitting at $\delta = 0$. Tail, volatility, and mean-dynamics regimes appear in separate developmental simulations, not in a completed isolated-factor robustness experiment.

5.3 Robustness estimands

For paired realization b , method m , comparator c , and metric M , define

$$D_b^{m,c}(\delta) = M_b^m(\delta) - M_b^c(\delta).$$

For a future experiment with B independent DGP realizations, the intended summary is

$$\overline{D}^{m,c}(\delta) = \frac{1}{B} \sum_{b=1}^B D_b^{m,c}(\delta), \quad \text{MCSE}(\overline{D}) = \frac{s_D(\delta)}{\sqrt{B}}.$$

Neural-seed repetitions on a prespecified subset separate DGP variability from optimization variability. Failure frequencies and runtimes are part of the estimand rather than discarded implementation details.

6 Experimental Design

6.1 Four-model replication matrix

The core experiment follows the comparison logic of [1]:

Table 1: Mapping from calibration-aware classification to forecasting.

Source-paper model	Forecasting model	Objective components
FNN	RNN	NLL
BNN	BRNN	expected NLL + KL
CA-FNN	CA-RNN	NLL + projected-PIT calibration
CA-BNN	CA-BRNN	expected NLL/calibration + KL

Architectures, histories, splits, optimizers, stopping rules, and predictive heads are matched wherever the Bayesian formulation permits. The same projection matrix is used for all models in a comparison.

6.2 Calibration-weight experiment

Analogously to the source paper’s plot of ECE and accuracy against λ , this study plots projected-PIT calibration error and predictive performance against λ_{cal} . The paired panels will contain:

1. projected calibration discrepancy and reliability curves;
2. Energy Score, NLL where comparable, RMSE, interval score, and width.

The result of interest is a Pareto relationship, not calibration alone. Separate curves are required for CA-RNN and CA-BRNN.

Table 2: Loss-component ablation.

Specification	NLL	$\mathcal{L}_{\text{cal}}$	$\mathcal{L}_{\text{seq}}$
RNN	✓		
CA-RNN	✓	✓	
CA-RNN (sequential)	✓	✓	✓

6.3 Sequential ablation

The minimum ablation is: The Bayesian versions repeat this comparison with the KL term always present. Additional ablations vary coordinate-only versus augmented projections, smoothing temperature, batch length, lag order, prior scale, and β .

To prevent checkpoint selection from masquerading as a calibration effect, the primary loss-component ablation uses identical initial parameters, chronological mini-batches, optimization budgets, and predictive random numbers, with early stopping disabled. A secondary operational comparison may use early stopping, but all methods must then be selected by the same validation NLL rule.

6.4 Controlled simulations

Development simulations include Gaussian, heavy-tailed, heteroskedastic, and nonlinear regimes. A future confirmatory experiment should use independently generated data for every regime, sample size, shift severity, and DGP seed, with the following fixed in advance:

- sample sizes and number of independent realizations;
- neural-seed subset and number of optimization repetitions;
- calibration and KL grids;
- projection directions and smoothing grid;
- primary metrics and failure definitions; and
- rules for early stopping and invalid numerical runs.

6.5 Financial application

The application uses adjusted daily prices for AAPL, JPM, XOM, and WMT beginning in 2007 and converts them to aligned log returns. The four assets provide a small, computationally manageable multivariate system with heterogeneous sector exposure. All splits are chronological. Development selected $\lambda_{\text{cal}} = 100$ and $\lambda_{\text{seq}} = 2000$ before the multi-seed and rolling-regime analyses. The focused frequentist comparison uses a 32-unit GRU, a 20-day history, a joint Gaussian head, 500 forecast draws, and validation-NLL early stopping.

Standardization is fitted on each training partition. Validation contexts carry the final 20 training observations, and test contexts carry the final 20 validation observations; every evaluation target therefore retains an available historical context without using future information. RMSE, Energy Score, interval width, and interval score are reported in training-standardized units. Coverage and PIT diagnostics are invariant to the componentwise affine transformation. The two penalty weights

were chosen after inspecting developmental simulation and ablation outcomes, rather than by a pristine validation-only or preregistered selection procedure.

The rolling analysis uses expanding estimation samples and four non-overlapping test periods: 2017–2019 (pre-pandemic), 2020–2021 (pandemic), 2022–2023 (inflation and monetary tightening), and 2024–2025 (recent). Each window has a strictly subsequent validation period and five matched optimization seeds. These labels describe calendar periods rather than uniquely identified causal regimes. Because the full panel was observed during development and the final two windows overlap the initial test period, this is temporal robustness evidence rather than a pristine preregistered confirmatory experiment.

6.6 Paired uncertainty design

For model m , comparator c , seed s , and forecast origin t , let $d_{s,t}^{m,c}$ be a paired additive-score difference. The estimand is

$$\hat{\Delta}^{m,c} = \frac{1}{S} \sum_{s=1}^S \frac{1}{T} \sum_{t=1}^T d_{s,t}^{m,c}.$$

The hierarchical bootstrap resamples optimization seeds and, within each selected seed, matched forecast origins in moving blocks of 20 trading days. Projected-PIT calibration, coverage error, and PIT dependence are recomputed in every bootstrap draw. The initial financial analysis uses 2,000 draws and each rolling regime uses 1,000. Raw coverage and width remain descriptive; the inferential coverage loss is

$$E_{\text{cov}} = |\hat{C}_{0.90} - 0.90|.$$

Negative candidate-minus-comparator differences favor the candidate for loss metrics. An interval containing zero is called unresolved, not evidence of equality. Inference conditions on the architecture, tuning path, projection set, assets, and observed calendar windows.

6.7 Metrics and figures

The tables report mean performance, paired differences, and uncertainty intervals. The diagnostic figures include:

1. calibration and predictive metrics versus λ_{cal} ;
2. nominal-versus-empirical coverage curves;
3. projected PIT histograms or CDF reliability curves;
4. PIT autocorrelation diagnostics;
5. degradation curves versus shift severity; and
6. a calibration–sharpness or calibration–Energy Score frontier.

7 Results

7.1 Implementation validation

The implementation tests cover tensor dimensions, covariance positive-definiteness, loss evaluation, forecasting, chronological data handling, and paired-bootstrap identities. Eighteen automated

tests pass. The notebooks also completed the simulation, shift, financial, optimization-seed, and rolling-regime workflows. Successful execution establishes implementation consistency but is not itself statistical validation.

7.2 Four-model comparison

The initial matched comparison showed that calibration-aware objectives alter the forecast distribution but do not yield uniform dominance. CA-RNN made only a small projected-calibration improvement over RNN, while CA-BRNN did not improve on BRNN. Bayesian averaging did not remove the calibration–score tradeoff. The frequentist RNN, CA-RNN, and sequential CA-RNN therefore form the focused empirical comparison; BRNN and CA-BRNN remain supporting replications of the source paper’s model matrix.

7.3 Calibration-weight sensitivity

The exploratory sweep over $\lambda_{\text{cal}} \in \{0, 1, 10, 50, 100, 250, 500\}$ displayed a Pareto tradeoff: projected calibration improved through the intermediate penalty range, whereas likelihood and proper-score performance deteriorated. The value 100 was retained as a developmental compromise rather than selected to maximize a final test result. Calibration improvement alone is therefore not interpreted as a better probabilistic forecast.

7.4 Sequential calibration ablation

Relative to marginal CA-RNN, the sequential specification consistently repaired predictive performance. In the ten-seed financial experiment it improved MSE, Energy Score, interval score, nominal-coverage error, and projected-PIT calibration for every seed. It widened intervals, but simultaneous coverage and interval-score improvements show that the result is not merely unpenalized widening. Improvements in PIT temporal dependence were smaller and generally unresolved. The sequential penalty is empirically important, but the evidence does not establish PIT independence.

7.5 Misspecification and distribution shift

The frozen-model experiment jointly increased innovation scale and common correlation from severity zero to one. All specifications deteriorated. At zero, RNN had Energy Score 1.2094, interval score 3.8581, and projected calibration error 0.001068, compared with 1.2124, 3.9251, and 0.001545 for sequential CA-RNN. At maximum severity, the ordering reversed: RNN obtained 2.3671, 10.3070, and 0.009370, while sequential CA-RNN obtained 2.3416, 10.0120, and 0.007728. This single-seed combined stress is developmental evidence rather than an isolated causal test of scale or dependence. Absolute levels and paired degradation are considered together because a worse nominal starting point can mechanically reduce measured deterioration.

7.6 Financial results

Table 3 reports averages over ten matched optimization seeds on the original financial test period. Sequential CA-RNN reduced projected-PIT calibration error relative to RNN for all ten seeds, from 0.001382 to 0.001185, or approximately 14%. Its Energy Score and RMSE were worse for all ten seeds. The interval-score difference was unresolved, with a 95% hierarchical interval of $[-0.0047, 0.0235]$, as was the nominal-coverage error difference.

Table 3: Mean financial performance over ten matched optimization seeds. Lower is better except for coverage, whose nominal target is 0.90.

Model	RMSE	Energy	Interval	Coverage	PIT calibration
RNN	0.9368	1.1394	4.0509	0.8725	0.001382
CA-RNN	0.9474	1.1510	4.0979	0.8650	0.001265
Sequential CA-RNN	0.9448	1.1478	4.0612	0.8748	0.001185

Plain CA-RNN isolates why the extension matters. Against RNN it had worse MSE, Energy Score, interval score, and coverage error despite lower projected-PIT error. Sequential CA-RNN then improved MSE by 0.00483, Energy Score by 0.00329, interval score by 0.03673, and coverage error by 0.00979 relative to CA-RNN; every corresponding hierarchical interval excluded zero. It repairs a substantial part, but not all, of the cost of marginal calibration.

7.7 Rolling market-regime results

Table 4 gives five-seed means. Sequential CA-RNN had lower projected-PIT calibration error than RNN in all four regimes. Hierarchical intervals favored it in the pre-pandemic, inflation/tightening, and recent windows; the pandemic difference was unresolved. In contrast, its Energy Score was lower only pre-pandemic, where the difference was unresolved.

Table 4: Five-seed regime means for RNN (R) and sequential CA-RNN (S).

Regime	Energy Score		Interval score		PIT calibration	
	R	S	R	S	R	S
Pre-pandemic	0.8629	0.8626	3.1123	3.1027	0.001914	0.001503
Pandemic	1.5016	1.5029	5.7371	5.6629	0.002995	0.002883
Inflation/tightening	1.2809	1.2884	4.5862	4.5396	0.001842	0.001465
Recent	1.0237	1.0289	3.6648	3.6935	0.001224	0.000904

The interval-score ordering is regime-dependent. Sequential CA-RNN significantly outperformed RNN during the pandemic and inflation/tightening periods, was unresolved pre-pandemic, and significantly underperformed recently. Coverage error similarly favored sequential CA-RNN pre-pandemic and during inflation/tightening, was unresolved during the pandemic, and favored RNN recently. These reversals directly answer the distribution-shift question.

Across regimes, sequential CA-RNN improved Energy Score and interval score over marginal CA-RNN directionally in four of four windows. These improvements were resolved in the pandemic, inflation/tightening, and recent windows, while the pre-pandemic differences were unresolved. Projected calibration improved over marginal CA-RNN in three of four windows.

7.8 Interval-score mechanism under temporal shift

To explain the regime reversal, the retained forecasts were analyzed without refitting. For a 90% interval $[L_{t,j}, U_{t,j}]$, the componentwise interval score decomposes exactly as

$$S_{t,j} = \underbrace{U_{t,j} - L_{t,j}}_{\text{width}} + \underbrace{20(L_{t,j} - Y_{t,j})\mathbf{1}\{Y_{t,j} < L_{t,j}\}}_{\text{lower-tail penalty}} + \underbrace{20(Y_{t,j} - U_{t,j})\mathbf{1}\{Y_{t,j} > U_{t,j}\}}_{\text{upper-tail penalty}}. \quad (12)$$

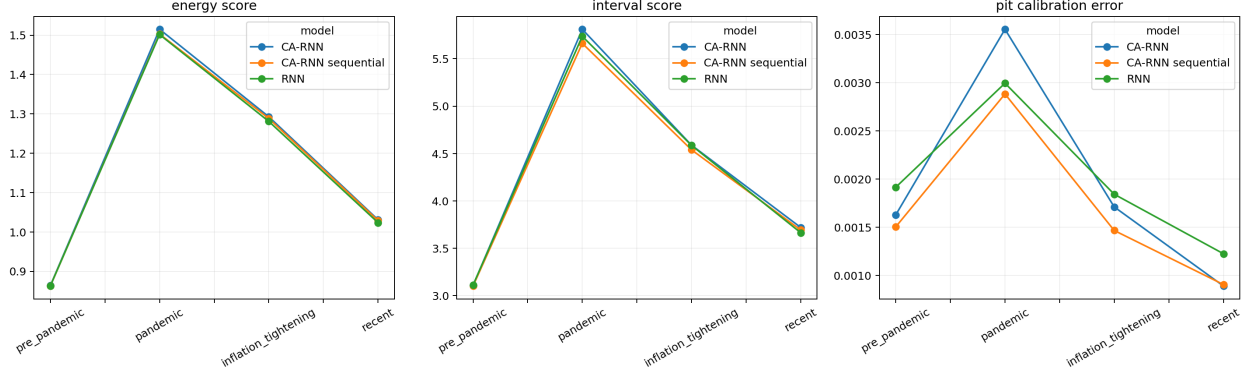


Figure 1: Mean Energy Score, interval score, and projected-PIT calibration error across five matched seeds and four chronological test regimes. Lower is better.

Table 5 reports the mean sequential-CA-RNN-minus-RNN difference in each component. The arithmetic identity in (12) means the final column is exactly the sum of the preceding three columns.

Table 5: Decomposition of the interval-score difference between sequential CA-RNN and RNN. Negative total differences favor sequential CA-RNN.

Regime	Width	Lower penalty	Upper penalty	Total
Pre-pandemic	−0.0439	+0.0241	+0.0103	−0.0096
Pandemic	+0.0420	−0.0599	−0.0563	−0.0742
Inflation/tightening	+0.2342	−0.1709	−0.1098	−0.0466
Recent	−0.0185	+0.0268	+0.0205	+0.0287

The pandemic and inflation/tightening improvements arise through beneficial widening: the added width is more than offset by reductions in both tail-miss penalties. In the recent period, sequential CA-RNN instead produces slightly narrower intervals and larger penalties in both tails. Before the pandemic, the mechanism is different again: reduced width outweighs the additional tail penalties, yielding a small and statistically unresolved sharpness benefit.

The stress-period effect is concentrated in forecasts preceded by elevated volatility. Sequential CA-RNN’s interval-score differences in the pandemic are +0.0198, −0.0725, and −0.1705 for low-, medium-, and high-volatility origins, respectively. During inflation/tightening they are +0.0209, −0.0562, and −0.1045. By contrast, the recent-period differences are positive in every state: +0.0318, +0.0238, and +0.0305. Volatility states use trailing 20-day realized volatility known before each forecast and within-regime terciles.

The improvements are also asset-concentrated. XOM accounts for most of the inflation/tightening gain, with an interval-score difference of −0.2364; AAPL, JPM, and WMT have differences of −0.0051, +0.0221, and +0.0331. Pandemic improvements occur for AAPL, XOM, and WMT but not JPM. Recent losses are concentrated in AAPL, XOM, and WMT, while JPM improves slightly. These results preclude interpreting an aggregate gain as uniform across assets.

The train-to-test descriptors support, but do not prove, a scale-and-tail interpretation. Pandemic volatility is 1.26 times its training level and the mean standardized fourth moment is 29.87. The corresponding values are 1.00 and 12.08 during inflation/tightening, versus 0.83 and 6.37 recently. The recent period has the largest standardized location shift and correlation shift of the four windows. Thus sequential calibration appears most useful for persistent scale and tail errors, while

lower-volatility location or dependence changes remain harder.

Importantly, projected and coordinate PIT calibration still improve recently even as 90% interval performance deteriorates. The calibration loss averages 19 CDF grid levels and multiple projections; it does not target the 5% and 95% quantiles specifically. Better global PIT uniformity can therefore coexist with worse tail coverage. The exact score decomposition is established by the retained forecasts, whereas the proposed economic mechanism is post-hoc and requires new data for confirmation.

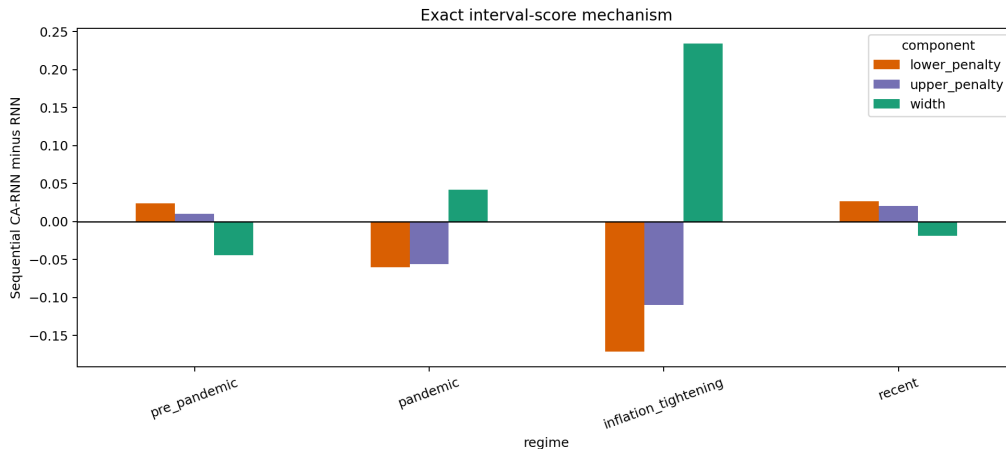


Figure 2: Exact decomposition of sequential-CA-RNN-minus-RNN interval-score differences. Tail reductions offset widening in the pandemic and inflation/tightening periods, but not in the recent period.

8 Discussion

8.1 Interpretation

Three conclusions follow. First, a marginal calibration penalty is insufficient for dependent multivariate forecasting: it can improve its targeted diagnostic while harming proper scores and nominal coverage. Second, the sequential loss is not cosmetic; it consistently improves marginal CA-RNN and recovers much of its predictive-performance loss. Third, sequential CA-RNN is a calibration-oriented method rather than a uniformly more accurate forecaster. Its projected-calibration advantage is stable across seeds and calendar periods, whereas its proper-score advantage depends on regime and scoring rule.

The disagreement between Energy Score and interval score is substantively useful. Energy Score assesses the multivariate predictive distribution, whereas the marginal interval score rewards coverage and sharpness asset by asset. During the pandemic and inflation/tightening periods, sequential CA-RNN improves the latter without improving the former. Its practical value therefore depends on whether an application prioritizes marginal risk intervals or overall multivariate accuracy.

8.2 Limitations

Limitations include finite projection sets, four assets, one-step horizons, sensitivity to smoothing and batching, mean-field variational misspecification, a Gaussian predictive head, and limited ability to establish conditional calibration from finite dependent sequences. Regime labels are calendar-based

and partly specified after broader exposure to the data. The hierarchical bootstrap represents seed and within-window sampling variation but conditions on the architecture, tuning path, assets, projections, and chosen windows. Moving-block resampling can distort PIT autocorrelation at artificial block boundaries, so temporal-dependence intervals are interpreted cautiously. Finally, improved finite-projection calibration does not imply full joint or conditional calibration.

8.3 Future work

The most direct methodological extension is a tail-aware sequential calibration objective. The current loss weights all 19 PIT grid levels and all fixed projections equally. A weighted version could emphasize application-relevant tail levels,

$$\mathcal{L}_{\text{tail}} = \frac{1}{R \sum_g w_g} \sum_{r=1}^R \sum_{g=1}^G w_g \left[\frac{1}{|B|} \sum_{t \in B} s_\tau(q_g - U_{t,r}) - q_g \right]^2, \quad (13)$$

with larger prespecified w_g near $q = 0.05$ and $q = 0.95$. A complementary extension would penalize lagged dependence in tail exceedance indicators rather than only covariance of centered PITs. Either extension must be tuned on validation data and evaluated with global calibration and proper scores to avoid improving one nominal interval at the expense of the rest of the distribution.

Further work should test whether the observed high-volatility benefit replicates for new assets, frequencies, forecast horizons, and genuinely held-out calendar periods. Conditional calibration diagnostics could use volatility, market return, and liquidity states fixed before evaluation. Heavy-tailed or mixture predictive heads may separate deficiencies in the Gaussian output law from deficiencies in the calibration objective. Finally, uncertainty for PIT dependence should use a procedure that avoids correlations introduced at moving-block boundaries.

9 Conclusion

This paper adapts calibration-aware Bayesian learning to continuous multivariate recurrent forecasting and introduces a temporal calibration penalty based on lagged projected-PIT covariance. Calibration-aware training reliably changes forecast calibration, but marginal regularization alone incurs predictive costs. The sequential extension consistently improves the marginal CA-RNN specification and produces lower projected-PIT calibration error than a likelihood-trained RNN across ten optimization seeds and four chronological market regimes.

The method is not uniformly superior. RNN retains better point and Energy Score performance in much of the financial evidence, and interval-score advantages reverse across regimes. The contribution is therefore not a dominance claim. It is evidence that temporal calibration must be addressed explicitly, that doing so repairs the direct calibration-aware adaptation, and that the remaining calibration–accuracy tradeoff varies systematically under temporal distribution shift.

A Implementation Correspondence

References

- [1] J. Huang, S. Park, and O. Simeone, “Calibration-Aware Bayesian Learning,” *arXiv preprint arXiv:2305.07504*, version 2, 2024.

Table 6: Mathematical objects and current source modules.

Object	Source location
GRU and joint Gaussian head	<code>src/innovcal/ca_rnn/model.py</code>
Mean-field variational layers	<code>src/innovcal/ca_rnn/layers.py</code>
Cholesky transformation	<code>src/innovcal/ca_rnn/distributions.py</code>
Projected PIT and calibration losses	<code>src/innovcal/ca_rnn/losses.py</code>
Training and early stopping	<code>src/innovcal/ca_rnn/trainer.py</code>
Predictive sampling	<code>src/innovcal/ca_rnn/predict.py</code>
Controlled DGPs	<code>src/innovcal/data/simulation.py</code>
Sample-based evaluation	<code>src/innovcal/evaluation/metrics.py</code>
Hierarchical paired inference	<code>src/innovcal/evaluation/uncertainty.py</code>
Seed and regime experiments	<code>src/innovcal/experiments/reproducibility.py</code>